

Scenario 1: Student Academic Performance Dashboard

Code :

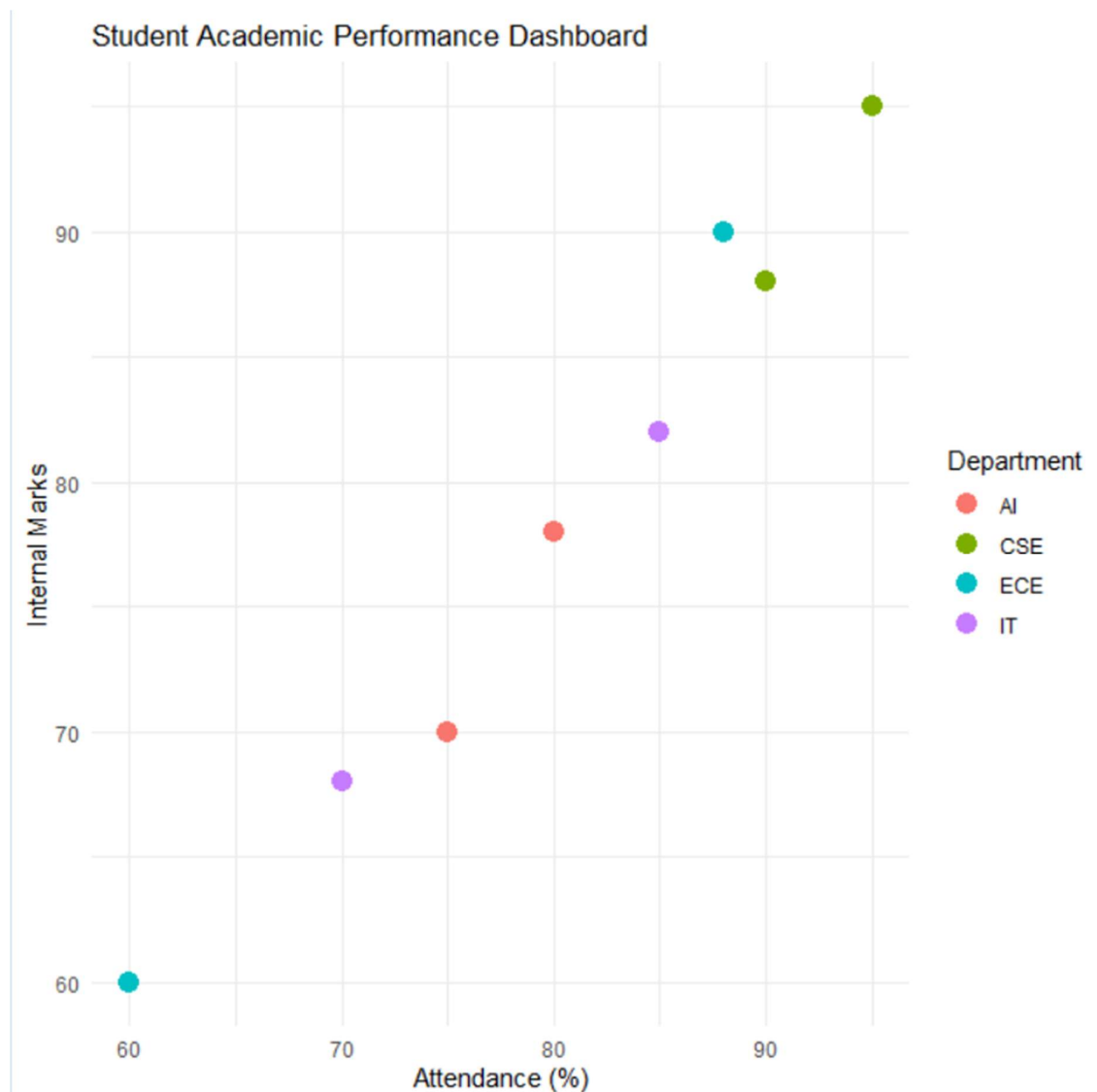
```
# Install ggplot2 if not already installed
# install.packages("ggplot2")

library(ggplot2)

student_data <- data.frame(
  Student_ID = c(101,102,103,104,105,106,107,108),
  Attendance = c(90,75,85,60,95,80,70,88),
  Internal_Marks = c(88,70,82,60,95,78,68,90),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE")
)

# Scatter Plot
ggplot(student_data,
  aes(x = Attendance,
      y = Internal_Marks,
      color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

Output:



Scenario 2: Electric Vehicle Market Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```

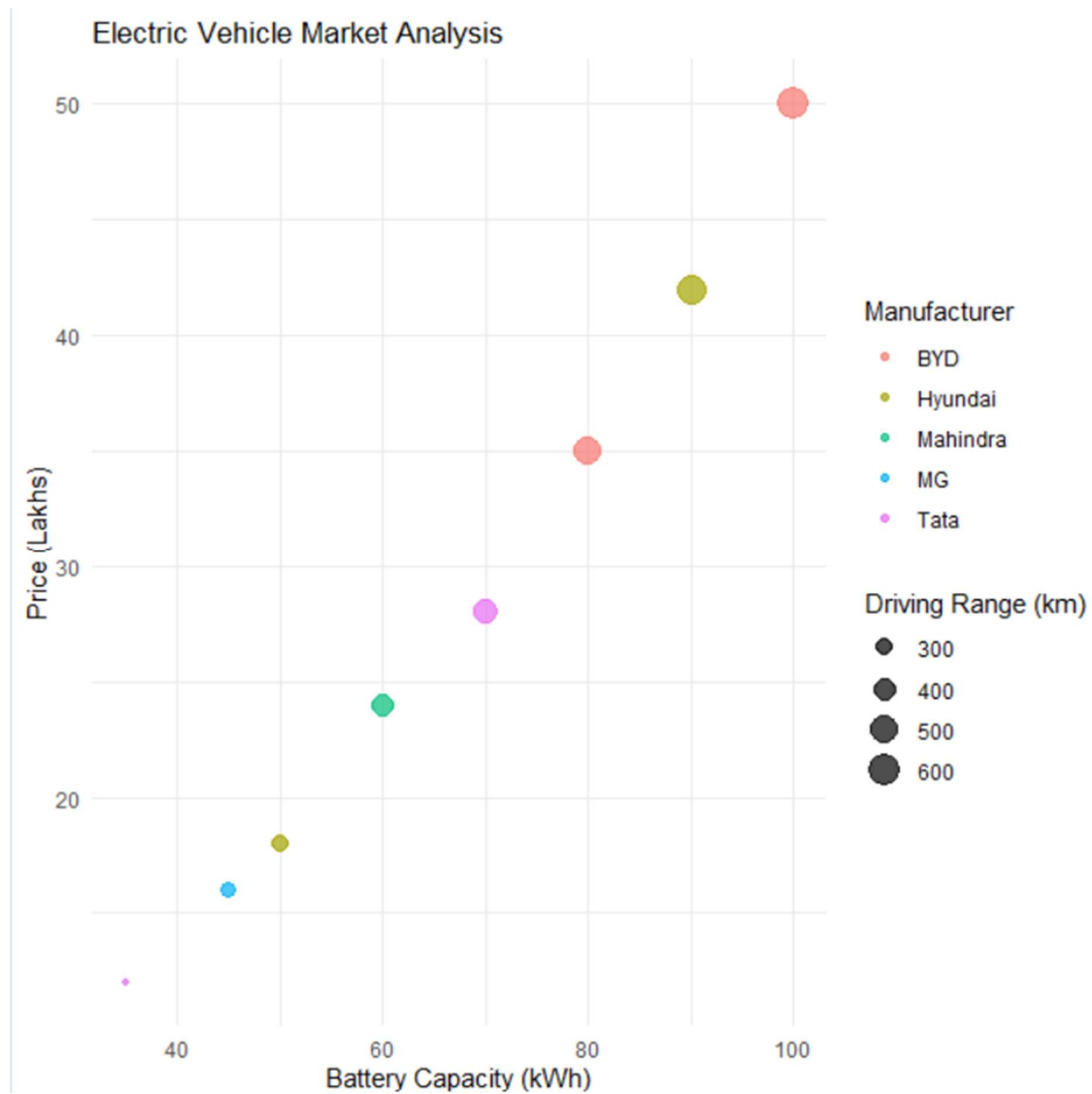
```

ev_data <- data.frame(
  Vehicle_Model = c("Model A", "Model B", "Model C", "Model D",
                    "Model E", "Model F", "Model G", "Model H"),
  Battery_Capacity = c(35, 45, 50, 60, 70, 80, 90, 100),
  Price = c(12, 16, 18, 24, 28, 35, 42, 50), # Price in Lakhs
  Driving_Range = c(220, 280, 300, 380, 420, 500, 550, 620),
  Manufacturer = c("Tata", "MG", "Hyundai", "Mahindra",
                   "Tata", "BYD", "Hyundai", "BYD")
)

# Bubble Chart
ggplot(ev_data,
  aes(x = Battery_Capacity,
      y = Price,
      size = Driving_Range,
      color = Manufacturer)) +
  geom_point(alpha = 0.7) +
  labs(
    title = "Electric Vehicle Market Analysis",
    x = "Battery Capacity (kWh)",
    y = "Price (Lakhs)",
    size = "Driving Range (km)",
    color = "Manufacturer"
  ) +
  theme_minimal()

```

Output:



Scenario 3: Hospital Patient Admission Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```

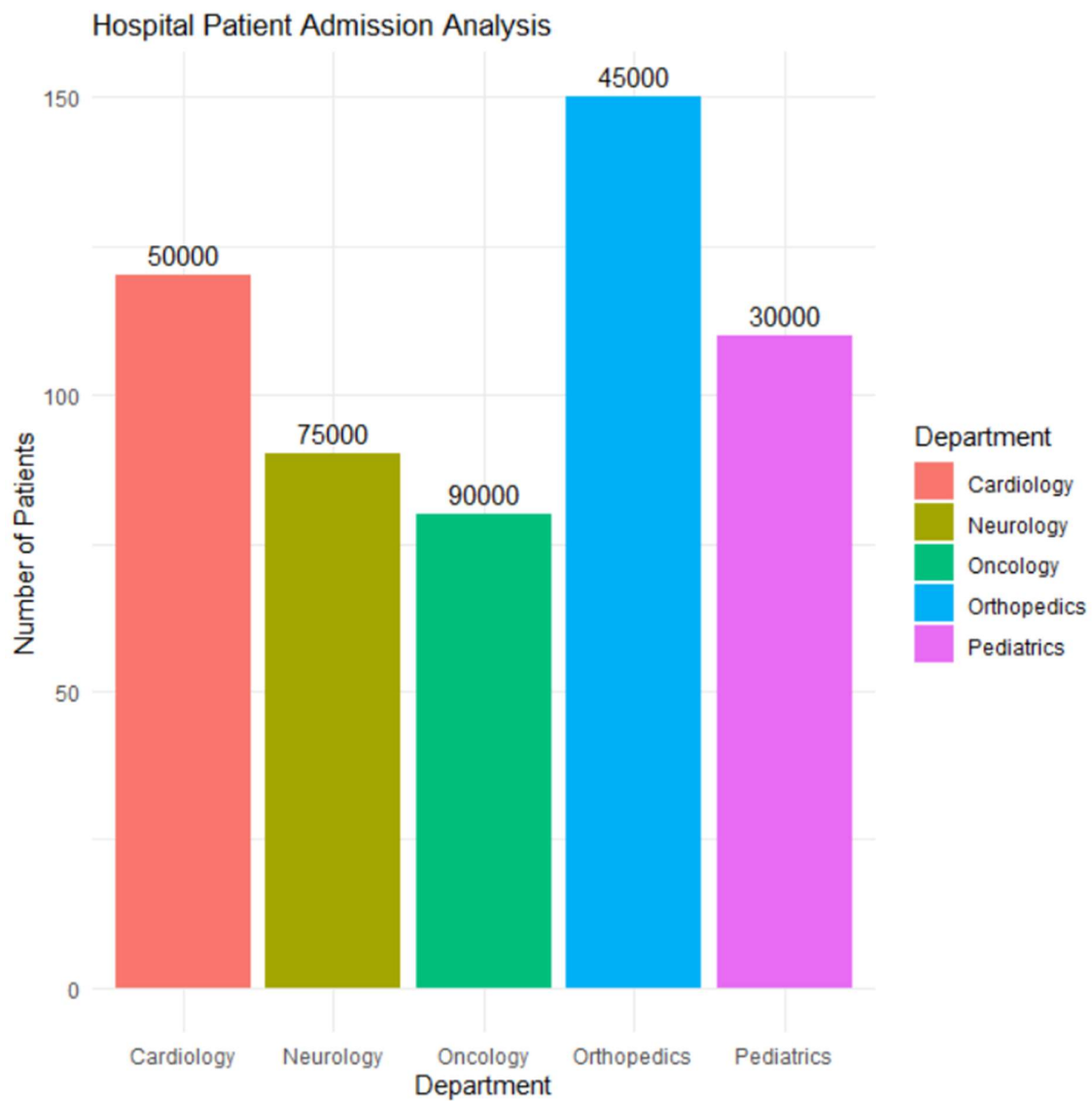
```

hospital_data <- data.frame(
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics", "Oncology"),
  Number_of_Patients = c(120, 90, 150, 110, 80),
  Average_Treatment_Cost = c(50000, 75000, 45000, 30000, 90000)
)

# Bar Chart
ggplot(hospital_data,
  aes(x = Department,
    y = Number_of_Patients,
    fill = Department)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Average_Treatment_Cost),
    vjust = -0.5,
    size = 4) +
  labs(
    title = "Hospital Patient Admission Analysis",
    x = "Department",
    y = "Number of Patients",
    fill = "Department"
  ) +
  theme_minimal()

```

Output:



Scenario 4: Weather Monitoring System

Code:

```
# Install ggplot2 if not installed
# install.packages("ggplot2")

library(ggplot2)
```

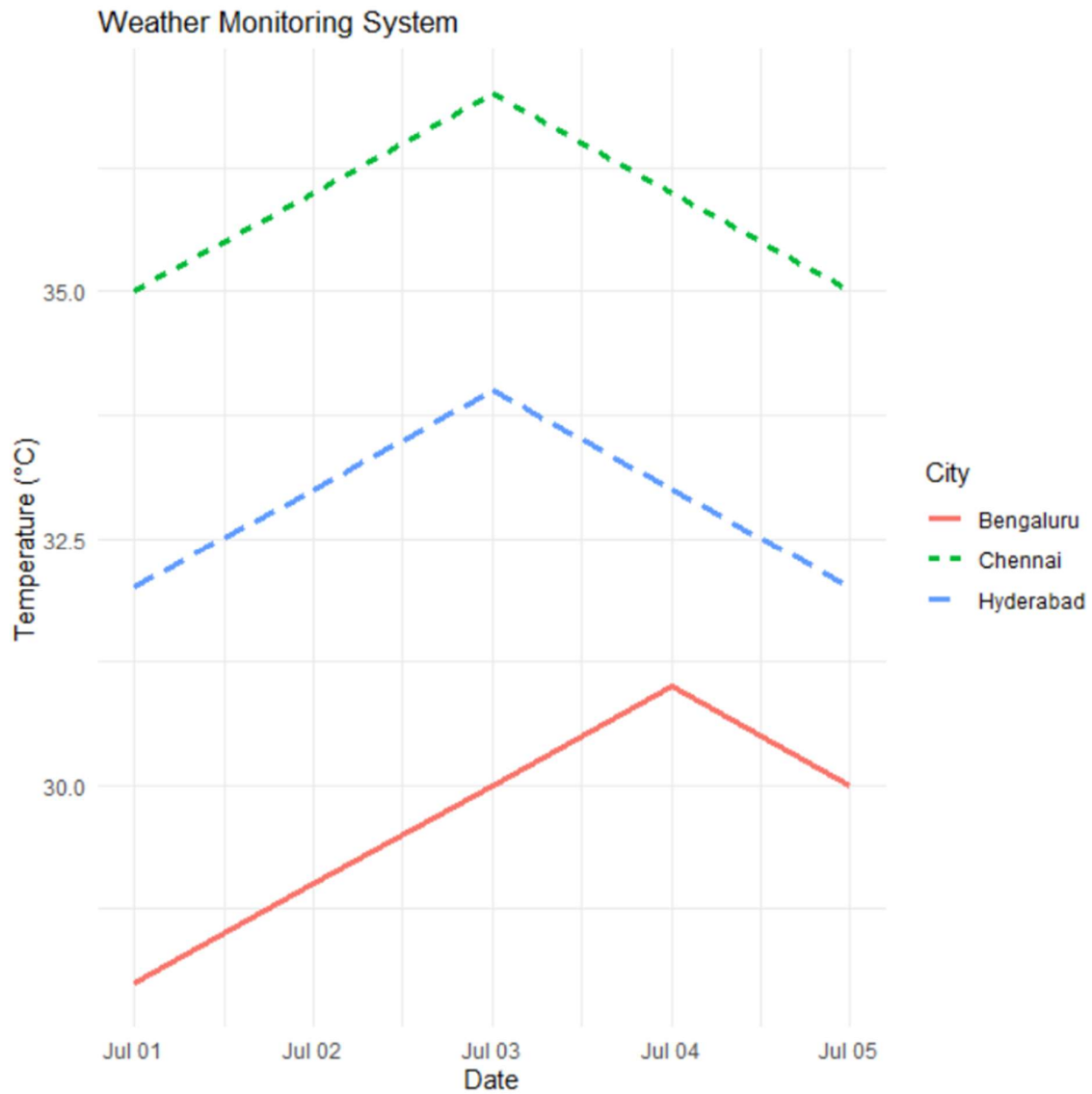
```

# Sample dataset
weather_data <- data.frame(
  Date = rep(as.Date(c("2025-07-01", "2025-07-02", "2025-07-03", "2025-07-04", "2025-07-05")), 3),
  Temperature = c(
    35, 36, 37, 36, 35, # Chennai
    28, 29, 30, 31, 30, # Bengaluru
    32, 33, 34, 33, 32 # Hyderabad
  ),
  City = rep(c("Chennai", "Bengaluru", "Hyderabad"), each = 5)
)

# Line Chart
ggplot(weather_data,
  aes(x = Date,
    y = Temperature,
    color = City,
    linetype = City,
    group = City)) +
  geom_line(size = 1.2) +
  labs(
    title = "Weather Monitoring System",
    x = "Date",
    y = "Temperature (°C)",
    color = "City",
    linetype = "City"
  ) +
  theme_minimal()

```

Output:



Scenario 5: Supermarket Sales Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```



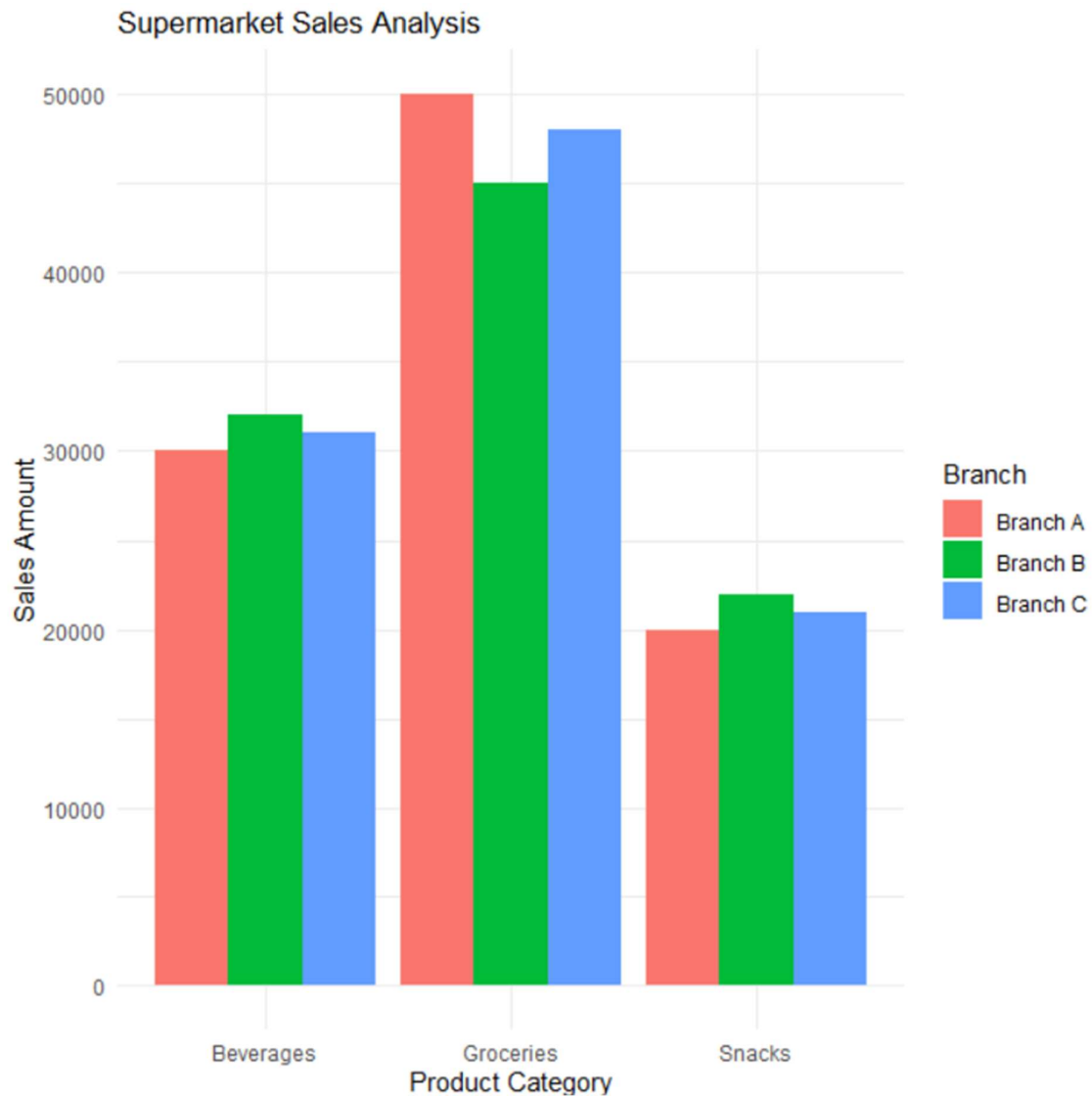
```

sales_data <- data.frame(
  Month = rep("January", 9),
  Product_Category = rep(c("Groceries", "Beverages", "Snacks"), each = 3),
  Sales_Amount = c(50000, 45000, 48000,
                   30000, 32000, 31000,
                   20000, 22000, 21000),
  Branch = rep(c("Branch A", "Branch B", "Branch C"), 3)
)

# Grouped Bar Chart
ggplot(sales_data,
       aes(x = Product_Category,
           y = Sales_Amount,
           fill = Branch)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Supermarket Sales Analysis",
    x = "Product Category",
    y = "Sales Amount",
    fill = "Branch"
  ) +
  theme_minimal()

```

Output:



Scenario 6:

Code:

```
library(ggplot2)
```

```
set.seed(123)
```

```
n <- 250
```

```
departments <- c("Engineering", "Sales", "HR", "Marketing", "Finance")
```

```
dataset <- data.frame(
```

```
  Employee_ID = 1000 + 1:n,
```

```
Department = sample(departments, n, replace = TRUE),  
Experience = round(runif(n, 1, 15), 1)  
)
```

```
dataset$Salary <- 40000 + (dataset$Experience * 4500) +  
  ifelse(dataset$Department == "Engineering", 15000, 0) +  
  rlnorm(n, meanlog = 9.5, sdlog = 0.5)
```

```
dataset$Salary <- round(dataset$Salary, -2)
```

```
plot_default_bins <- ggplot(dataset, aes(x = Salary)) +  
  geom_histogram(bins = 30, fill = "#2b5c8f", color = "white", alpha = 0.85) +  
  labs(  
    title = "Standard Binning (Bins = 30)",  
    subtitle = "Default exploration of the overall distribution shape",  
    x = "Salary ($)",  
    y = "Employee Count"  
  ) +  
  theme_minimal(base_size = 11) +  
  theme(  
    plot.title = element_text(face = "bold", color = "#1a1a1a"),  
    plot.subtitle = element_text(color = "gray40"),  
    panel.grid.minor = element_blank(),  
    panel.grid.major.x = element_blank()  
  )
```

```
plot_fine_bins <- ggplot(dataset, aes(x = Salary)) +  
  geom_histogram(binwidth = 5000, fill = "#4682b4", color = "white", alpha = 0.85) +  
  labs(  
    title = "Fine Binning (Binwidth = $5,000)",
```

```

    subtitle = "Detailed views reveal local spikes and gaps in pay scales",
    x = "Salary ($)",
    y = "Employee Count"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", color = "#1a1a1a"),
    plot.subtitle = element_text(color = "gray40"),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
  )

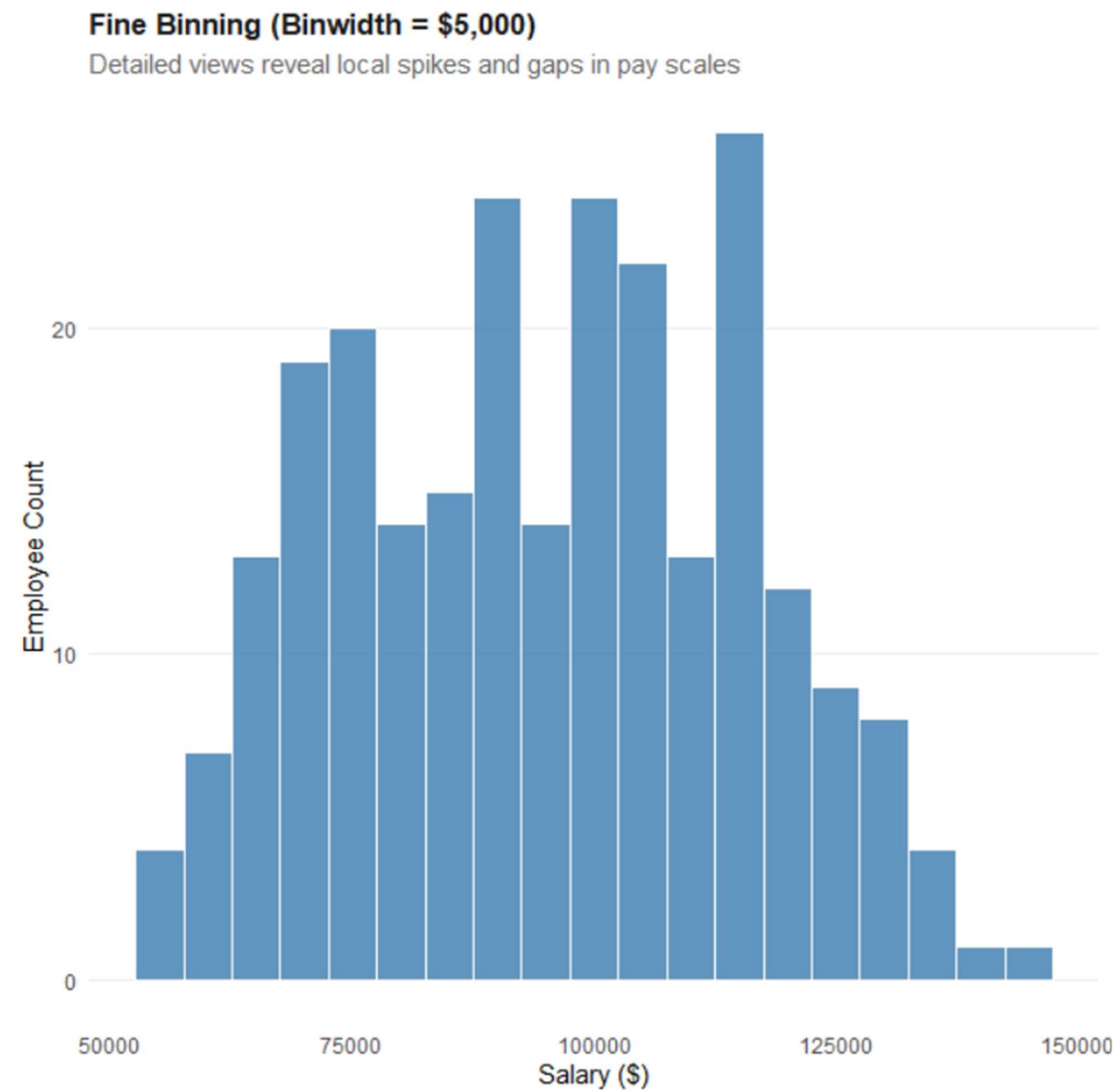
plot_log_scale <- ggplot(dataset, aes(x = Salary)) +
  geom_histogram(bins = 30, fill = "#e67e22", color = "white", alpha = 0.85) +
  scale_x_log10(labels = function(x) sprintf("$%sk", x/1000)) +
  labs(
    title = "Logarithmic Scale Transformation",
    subtitle = "Normalizes right-skewed extreme outliers for comparison",
    x = "Salary (Log Scale)",
    y = "Employee Count"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", color = "#1a1a1a"),
    plot.subtitle = element_text(color = "gray40"),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
  )

print(plot_default_bins)

```

```
print(plot_fine_bins)
print(plot_log_scale)
```

Output:



Scenario 7:

Code:

```
library(ggplot2)

brands <- c("Apple", "Samsung", "Google", "OnePlus", "Xiaomi")
set.seed(42)
```

```
n <- 100
```

```
dataset <- data.frame(  
  Brand = sample(brands, n, replace = TRUE),  
  RAM = sample(c(4, 6, 8, 12, 16), n, replace = TRUE),  
  Processor_Speed = runif(n, 1.8, 3.5),  
  Storage = sample(c(64, 128, 256, 512, 1024), n, replace = TRUE)  
)  
  
dataset$Price <- dataset$RAM * 50 + dataset$Processor_Speed * 150 + dataset$Storage * 0.5  
+ rnorm(n, 0, 50)
```

```
ggplot(dataset, aes(x = factor(RAM), y = Price, fill = Processor_Speed, size = Storage, shape  
= Brand)) +
```

```
  geom_point(position = position_jitter(width = 0.2, height = 0, seed = 42), alpha = 0.85,  
color = "#2b2b2b") +
```

```
  scale_shape_manual(values = c(21, 22, 23, 24, 25)) +
```

```
  scale_fill_gradient2(low = "#2c7bb6", mid = "#ffffbf", high = "#d7191c", midpoint = 2.65)  
+
```

```
  scale_size_continuous(range = c(3, 9), breaks = c(64, 128, 256, 512, 1024)) +
```

```
  labs(  
    x = "RAM (GB)",  
    y = "Price ($)",  
    shape = "Brand",  
    size = "Storage (GB)",  
    fill = "Processor Speed (GHz)",  
    title = "Smartphone Market Comparison",  
    subtitle = "Analysis of RAM and Processor Speed on Pricing Strategy"  
  ) +
```

```
  theme_minimal(base_size = 13) +
```

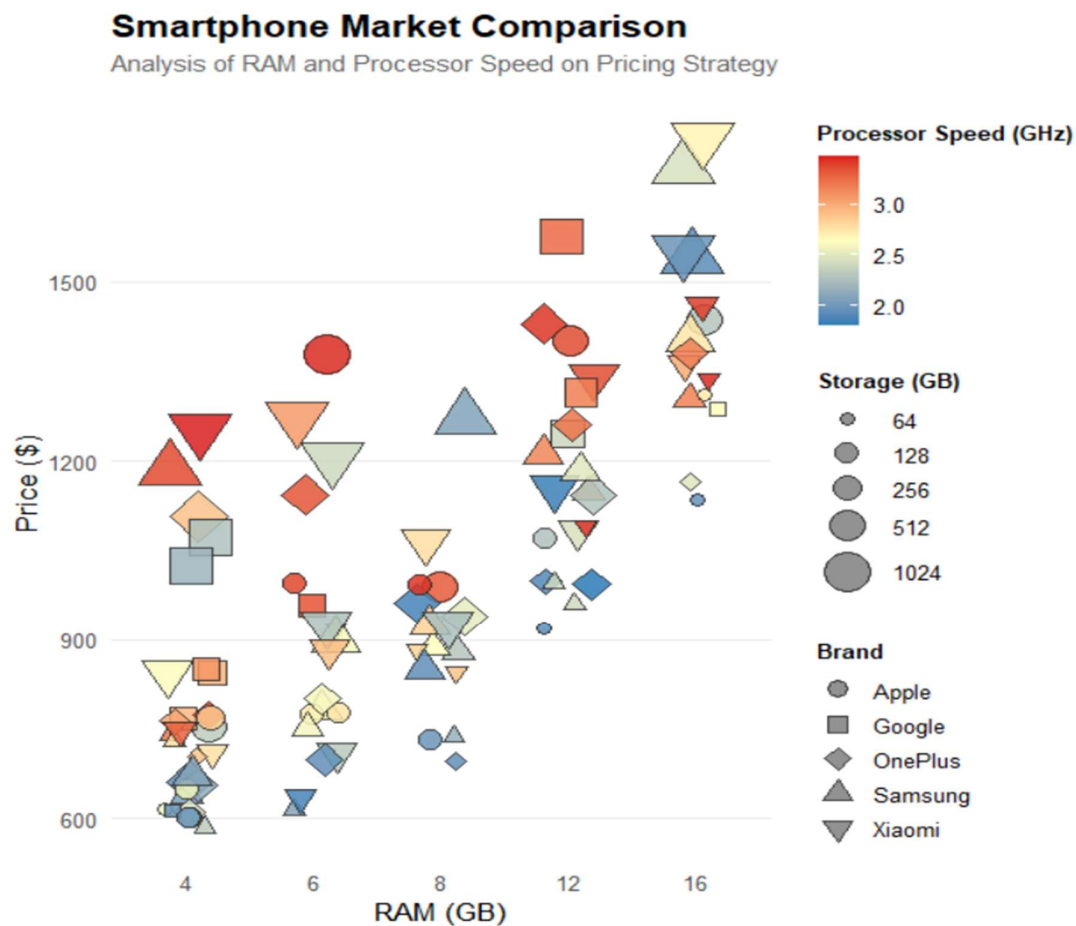
```
  theme(  
    plot.title = element_text(face = "bold", size = 16, margin = margin(b = 5)),  
    plot.subtitle = element_text(color = "gray40", size = 11, margin = margin(b = 15)),
```

```

panel.grid.major.x = element_blank(),
panel.grid.minor = element_blank(),
panel.grid.major.y = element_line(color = "gray92"),
legend.position = "right",
legend.title = element_text(face = "bold", size = 10),
legend.spacing.y = unit(0.3, "cm")
) +
guides(
  fill = guide_colorbar(order = 1, barheight = unit(3, "cm")),
  size = guide_legend(order = 2, override.aes = list(shape = 21, fill = "gray50")),
  shape = guide_legend(order = 3, override.aes = list(size = 4, fill = "gray50"))
)

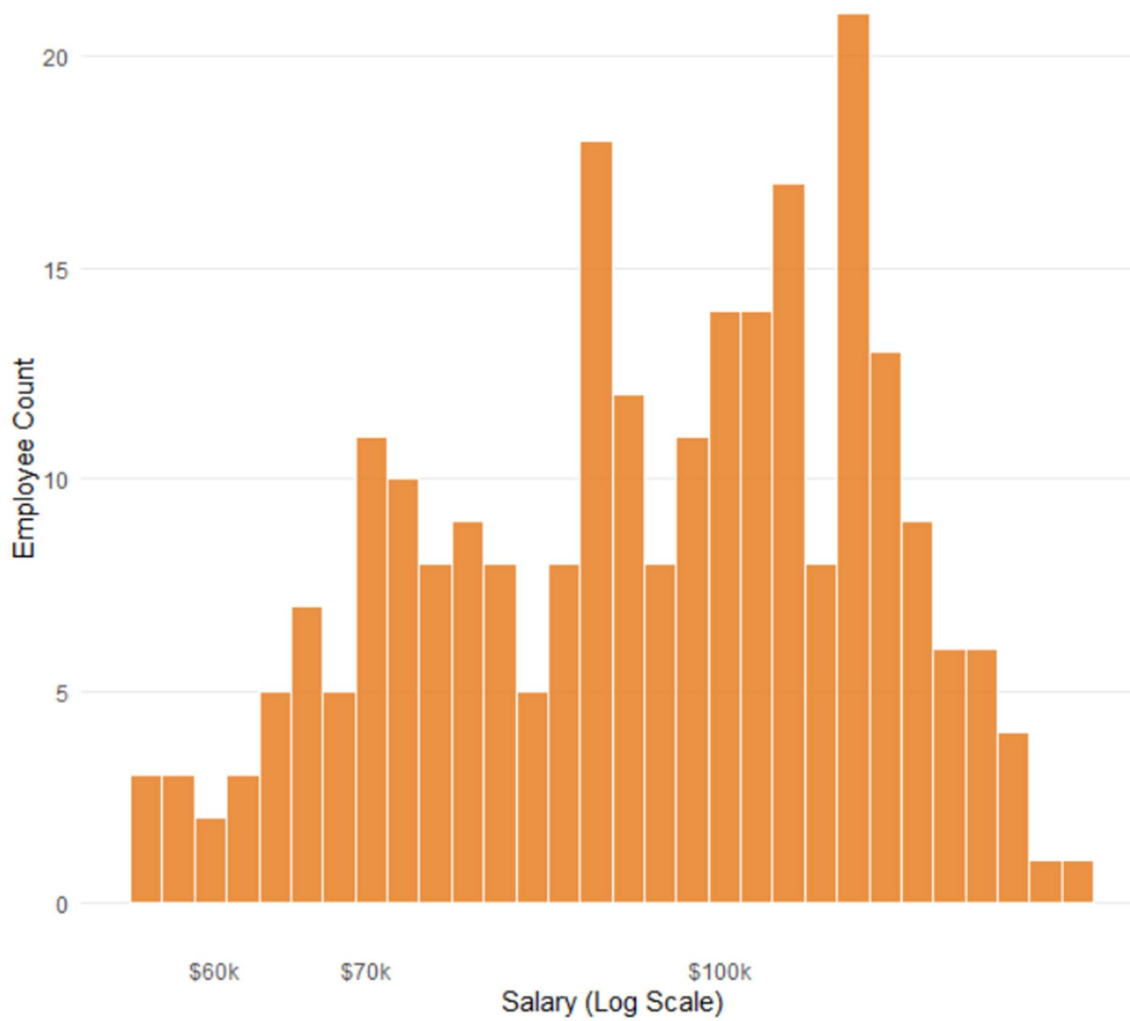
```

Output:



Logarithmic Scale Transformation

Normalizes right-skewed extreme outliers for comparison



Scenario 8:

Code:

```
library(ggplot2)
```

```
genres <- c("Action", "Comedy", "Drama", "Sci-Fi", "Romance", "Horror", "Thriller",  
"Documentary")
```

```
years <- 2021:2025
```

```
set.seed(88)
```

```
dataset <- expand.grid(Genre = genres, Release_Year = years)
```



```

dataset$Average_Rating <- runif(nrow(dataset), 5.5, 8.8)
dataset$Number_Of_Reviews <- sample(1000:50000, nrow(dataset), replace = TRUE)

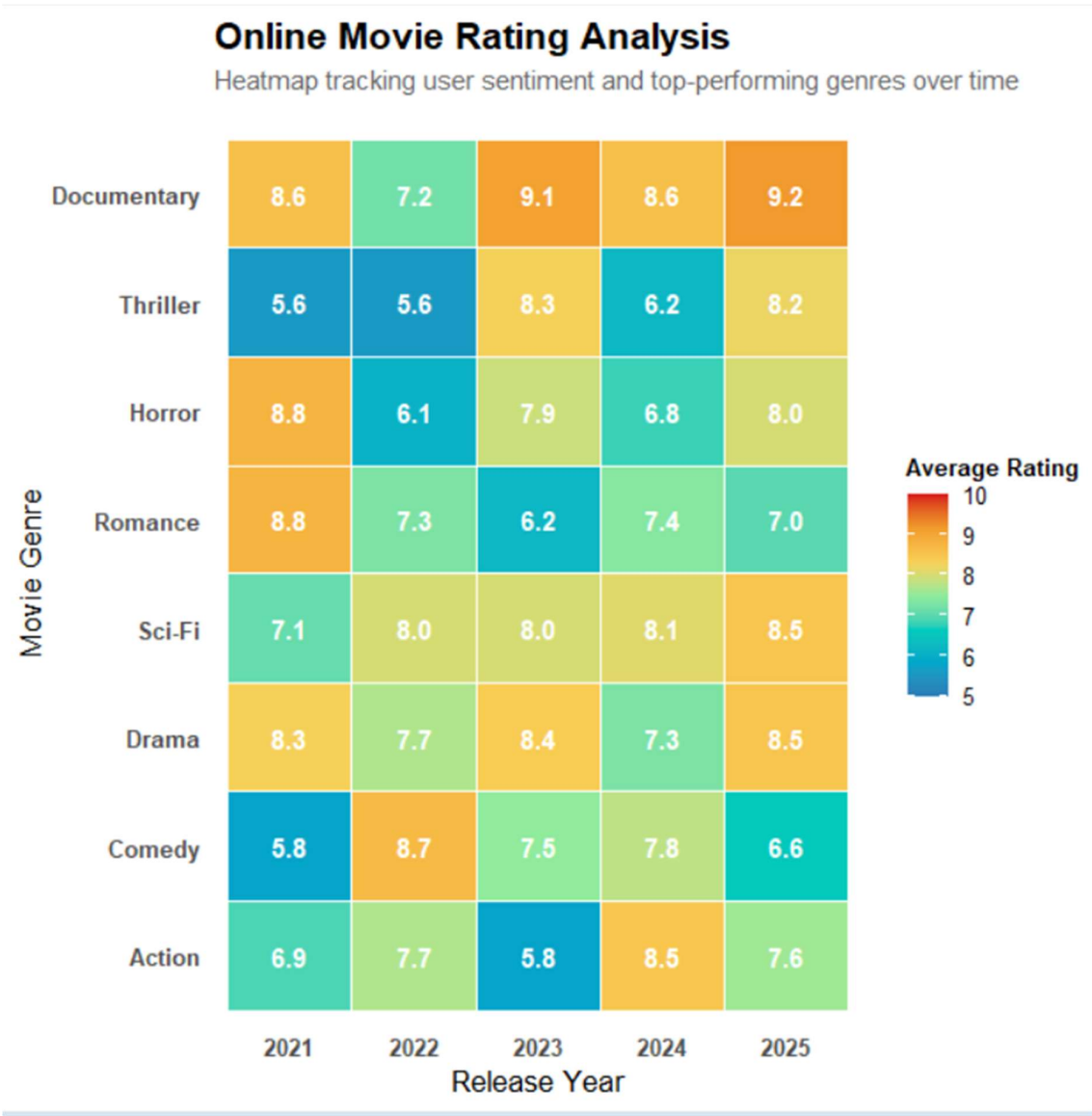
dataset$Average_Rating[dataset$Genre == "Documentary"] <-
dataset$Average_Rating[dataset$Genre == "Documentary"] + 0.6
dataset$Average_Rating[dataset$Genre == "Drama"] <-
dataset$Average_Rating[dataset$Genre == "Drama"] + 0.4
dataset$Average_Rating <- pmin(dataset$Average_Rating, 10)
dataset$Average_Rating <- round(dataset$Average_Rating, 1)

ggplot(dataset, aes(x = factor(Release_Year), y = Genre, fill = Average_Rating)) +
  geom_tile(color = "white", linewidth = 0.6) +
  geom_text(aes(label = sprintf("%.1f", Average_Rating)), color = "white", fontface = "bold",
size = 4) +
  scale_fill_gradientn(
    colors = c("#2c7bb6", "#00a6ca", "#00ccbc", "#90eb9d", "#f9d057", "#f29e2e",
"#d7191c"),
    limits = c(5, 10)
  ) +
  labs(
    x = "Release Year",
    y = "Movie Genre",
    fill = "Average Rating",
    title = "Online Movie Rating Analysis",
    subtitle = "Heatmap tracking user sentiment and top-performing genres over time"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16, margin = margin(b = 5)),
    plot.subtitle = element_text(color = "gray40", size = 11, margin = margin(b = 15)),
    panel.grid = element_blank(),
    axis.text.x = element_text(face = "bold"),

```

```
axis.text.y = element_text(face = "bold"),
legend.position = "right",
legend.title = element_text(face = "bold", size = 10),
legend.height = unit(5, "cm")
)
```

Output:



Scenario 9:

Code:

Output:

```
library(ggplot2)

states <- c("Maharashtra", "Kerala", "Karnataka", "Tamil Nadu", "Andhra Pradesh",
           "Uttar Pradesh", "Delhi", "West Bengal", "Odisha", "Rajasthan")

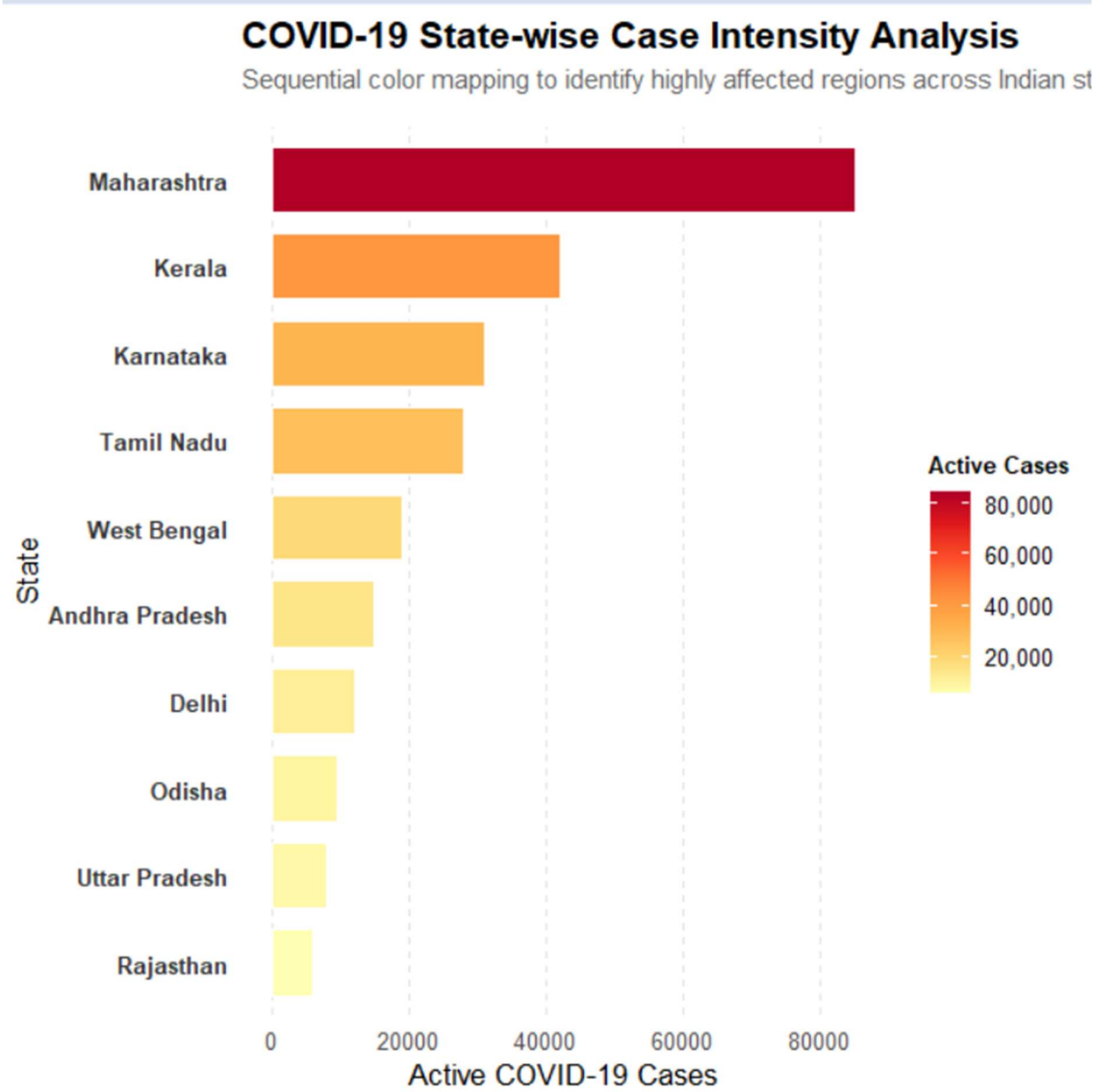
set.seed(77)

dataset <- data.frame(
  State = states,
  Active_Cases = c(85000, 42000, 31000, 28000, 15000, 8000, 12000, 19000, 9500, 6000),
  Vaccination_Percentage = runif(length(states), 65, 95),
  Population = c(123, 35, 68, 78, 53, 230, 20, 99, 46, 81)
)

ggplot(dataset, aes(x = reorder(State, Active_Cases), y = Active_Cases, fill = Active_Cases))
+
  geom_bar(stat = "identity", width = 0.75, color = "#ffffff", linewidth = 0.2) +
  coord_flip() +
  scale_fill_distiller(palette = "YlOrRd", direction = 1, labels = scales::comma) +
  labs(
    x = "State",
    y = "Active COVID-19 Cases",
    fill = "Active Cases",
    title = "COVID-19 State-wise Case Intensity Analysis",
    subtitle = "Sequential color mapping to identify highly affected regions across Indian
states"
  ) +
  theme_minimal(base_size = 13) +
```

```
theme(  
  plot.title = element_text(face = "bold", size = 16, margin = margin(b = 5)),  
  plot.subtitle = element_text(color = "gray40", size = 11, margin = margin(b = 15)),  
  panel.grid.major.y = element_blank(),  
  panel.grid.minor = element_blank(),  
  panel.grid.major.x = element_line(color = "gray90", linetype = "dashed"),  
  axis.text.y = element_text(face = "bold", color = "#333333"),  
  legend.position = "right",  
  legend.title = element_text(face = "bold", size = 10),  
  legend.height = unit(5, "cm")  
)
```

Output:



Scenario 10:

Code:

```
library(ggplot2)
```

```
data(iris)
```

```
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, color = Species, shape = Species, size =  
Sepal.Width)) +
```

```
  geom_point(alpha = 0.8) +
```

```

scale_shape_manual(values = c(16, 17, 18)) +
scale_color_manual(values = c("#e41a1c", "#377eb8", "#4daf4a")) +
scale_size_continuous(range = c(2, 6)) +
labs(
  x = "Sepal Length (cm)",
  y = "Petal Length (cm)",
  color = "Species",
  shape = "Species",
  size = "Sepal Width (cm)",
  title = "Iris Flower Classification Analysis",
  subtitle = "Morphological clustering using color, shape, and size dimensions"
) +
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(face = "bold", size = 16, margin = margin(b = 5)),
  plot.subtitle = element_text(color = "gray40", size = 11, margin = margin(b = 15)),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray95"),
  legend.position = "right",
  legend.title = element_text(face = "bold", size = 10)
) +
guides(
  color = guide_legend(order = 1),
  shape = guide_legend(order = 1),
  size = guide_legend(order = 2)
)

```

Output:

Iris Flower Classification Analysis

Morphological clustering using color, shape, and size dimensions

